UFR de mathématique et d'informatique
Université de Strasbourg

Eye Kalman Project

Pierre-Antoine Senger

Supervised by: Christophe Prud'homme, Thomas Saigre and Marcela Szopos

August 25, 2025

Overview

Context

The Human Eye

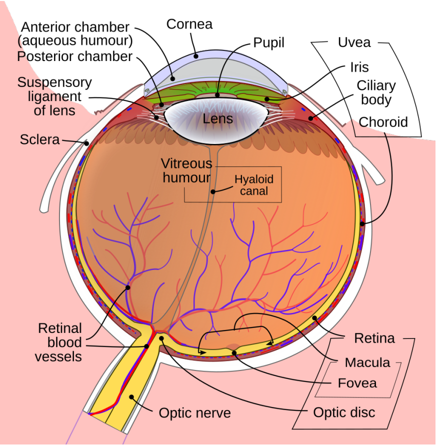


Figure 1: *[Schematic diagram of the human eye. Wikipedia]*

Light-Tissue Interactions

Coefficients of interest: **absorption** $\boxed{\mu_{a,i}(\lambda)}$, **scattering** $\boxed{\mu_{s,i}(\lambda)}$.

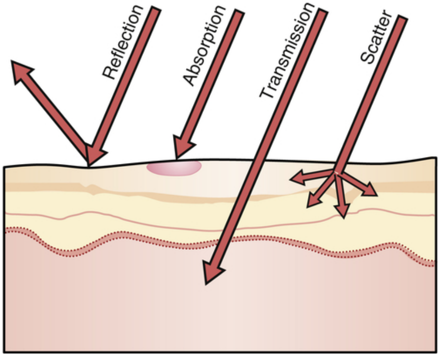


Figure 2: Laser tissue interactions. *["Laser Tissue Interactions: Biological Factors to Consider for Dermatology [Blo19].]*

Classification of Anterior Segment Laser Surgery (Part 1)

Laser Surgery	Wavelength	Pulse Duration	Target	Treatment
LASIK (Laser-Assisted in Situ Keratomileusis)	$1053\ \mathrm{nm}$ + $193\ \mathrm{nm}$	200–500 fs + continuous	Corneal stroma (under flap)	Myopia, hyperopia, astigmatism
LASEK (Laser-Assisted Subepithelial Keratectomy)	$193\ \mathrm{nm}$	Continuous	Superficial corneal stroma	Thin corneas, contact sports
SMILE (Small Incision Lenticule Extraction)	$1043\ \mathrm{nm}$	200–300 fs	Corneal stroma (lenticule)	Myopia, astigmatism
PRK (Photorefractive Keratectomy)	$193\ \mathrm{nm}$	Continuous	Corneal stroma (surface ablation)	Myopia, hyperopia, astigmatism
RLE (Refractive Lens Exchange)	1030–1053 nm	300–800 fs	Natural lens	Presbyopia, cataracts, high ametropia
Phakic IOL Implantation	—	—	Posterior/anterior chamber	High myopia without presbyopia

Classification of Anterior Segment Laser Surgery (Part 2)

Laser Surgery	Wavelength	Pulse Duration	Target	Treatment
Laser Blended Vision (LBV)	193 nm or fs	Continuous or fs	Both eyes (corneal stroma)	Presbyopia (monovision)
FLACS (Femtosecond Laser-Assisted Cataract Surgery)	1030–1053 nm	300–800 fs	Anterior capsule, lens	Cataract fragmentation
LPI (Laser Peripheral Iridotomy)	532 nm or 1064 nm	4–10 ns	Iris root	Narrow-angle glaucoma
YAG Capsulotomy (Posterior)	1064 nm	1–10 ns	Posterior capsule	Secondary cataract (PCO)
SLT (Selective Laser Trabeculoplasty)	532 nm	3 ns	Trabecular meshwork	Open-angle glaucoma

Source: https://www.discovervision.com/blog/common-types-of-laser-eye-surgery/

Femtosecond Laser (10^{-15} s)

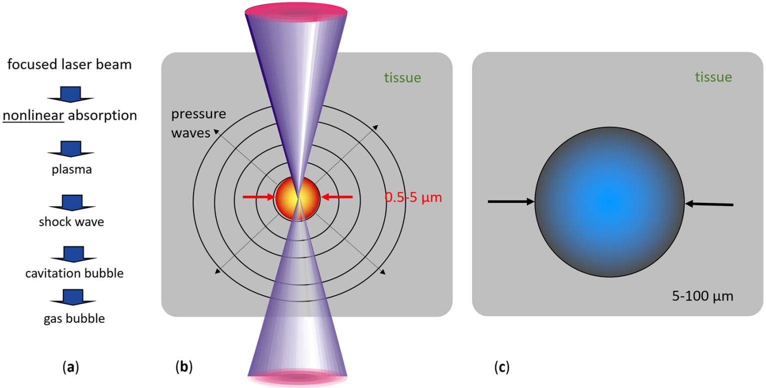


Figure 3: fs pulse laser effects in tissue: (a) sequence of events, (b) plasma diameter range (red) and emitted pressure wave pattern (circles), and (c) range of possible cavitation bubble dimensions (pulse energy dependent). *["Femtosecond lasers for eye surgery applications: historical overview and modern low pulse energy concepts" Asshauer et al.]*

Classification of Retinal Laser Surgery

Laser Surgery	Wavelength	Pulse Duration	Target	Treatment
Transpupillary Thermotherapy (TTT)	$810\ \mathrm{nm}$	$60\ \mathrm{s}$	Choroid	Choroidal melanoma, hemangioma, retinoblastoma
Panretinal Photocoagulation (PRP)	$532/577\ \mathrm{nm}$	$50–200\ \mathrm{ms}$	Entire retina	Diabetic retinopathy, neovascularization
Subthreshold Diode Micropulse (SDM)	$532/810\ \mathrm{nm}$	$100\ \mathrm{\mu s}$	RPE ($10\ \mathrm{\mu m}$)	Diabetic macular edema, CSCR, inherited retinopathies
Selective Retina Therapy (SRT)	$532\ \mathrm{nm}$	$1–10\ \mathrm{ns}$	Melanin in RPE ($2\ \mathrm{\mu m}$)	Macular edema, AMD, CSCR
Focal/Grid Photocoagulation	$532\ \mathrm{nm}$	$50–100\ \mathrm{ms}$	Localized retina	Microaneurysms in diabetic macular edema

Source: https://www.discovervision.com/blog/common-types-of-laser-eye-surgery/

Objectives

Project Objectives

- **Develop a mathematical model** coupling:
  - Fluence rate $\boxed{\Phi}$ $[\mathrm{W/m^2}]$: total radiant power per unit area from all directions.
  - Temperature $\boxed{T}$ $[K \text{ or } {}^\circ\mathrm{C}]$: thermal response in tissue.
- Build a realistic 3D geometric model of the eye.
- Solve the direct problem:
  Predict temperature distribution from known optical properties.
- Tackle the inverse problem using the Ensemble Kalman Filter (EnKF):
  Estimate tissue optical properties:
  - $\boxed{\mu_a(\lambda)}$: absorption coefficient
  - $\boxed{\mu_s(\lambda)}$: scattering coefficient
- Visualize and analyze the spatiotemporal results.
- Conduct a sensitivity analysis to identify dominant parameters.

Parameters Estimation

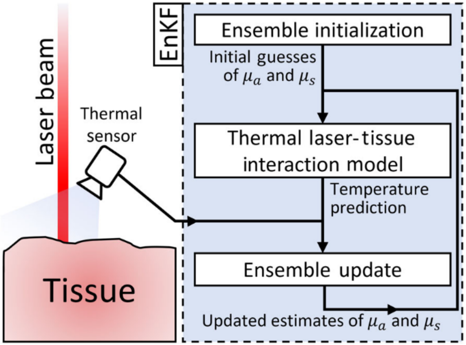


Figure 4: Laser-tissue interaction. *["Identification of tissue optical properties during thermal laser-tissue interactions: An ensemble Kalman filter-based approach." Arnold et al [AF22].]*

Laser 101

Laser Types

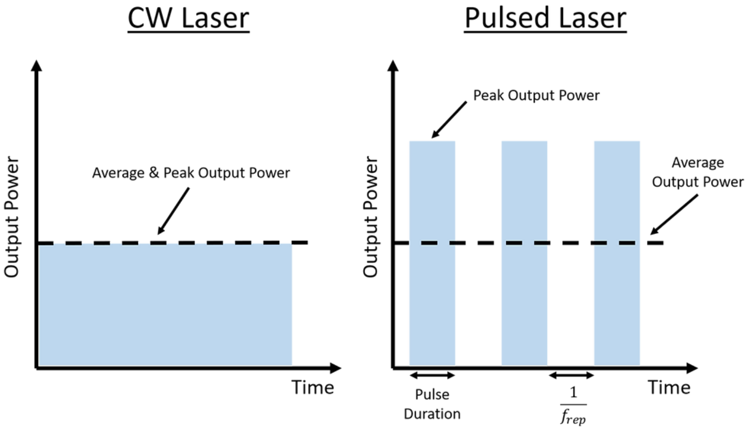


Figure 5: Continuous Wave (CW) vs Pulsed lasers.
[https://www.laser-protect.fr/en/post/laser-operating-modes-continuous-and-pulsed]

Key Parameters

- **Pulse Duration** τ: Ranges from fs (ultrashort) to ms (long-pulse).
- **Wavelength** λ: Distance between wave peaks.
  Optical properties depend on λ: $\boxed{\mu_a(\lambda)}$ = absorption, $\boxed{\mu_s(\lambda)}$ = scattering.
- **Laser Power** P: Typically 0.1 W–2 W.
- **Beam Waist** w_0: The tightest focus of the beam — where intensity peaks.
- **Beam Divergence** θ: Angular spread of the beam, in rad or mrad. For Gaussian beams:

$$\theta = \frac{\lambda}{\pi w_0}$$

- **Rayleigh Range** z_R: Distance from the waist to where beam cross-section doubles:

$$z_R = \frac{\pi w_0^2}{\lambda}$$

Beam Profiles

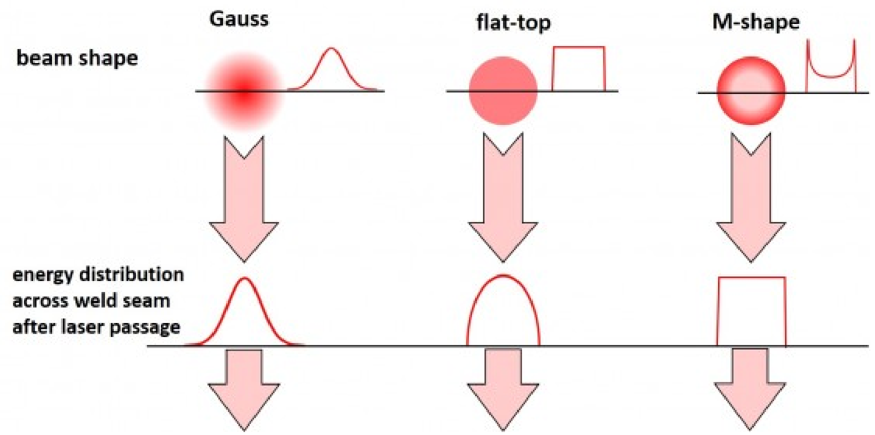


Figure 6: Different beam profiles. *["Beam Profiles" by Probylas]*

Pattern Scanning Laser

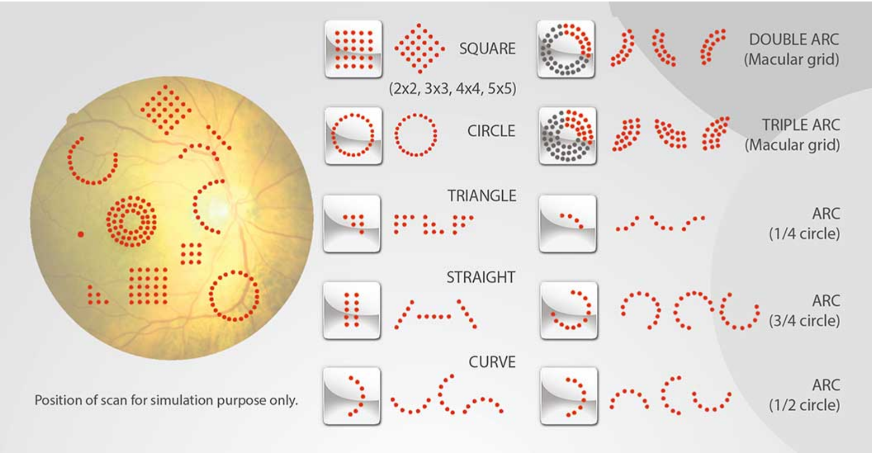


Figure 7: Pattern scanning laser. *["Muster Scan Laser" by Aeon Meditec]*

Mathematical Model

Eye Geometry $\Omega = \bigcup_{i=1}^{10} \Omega_i$

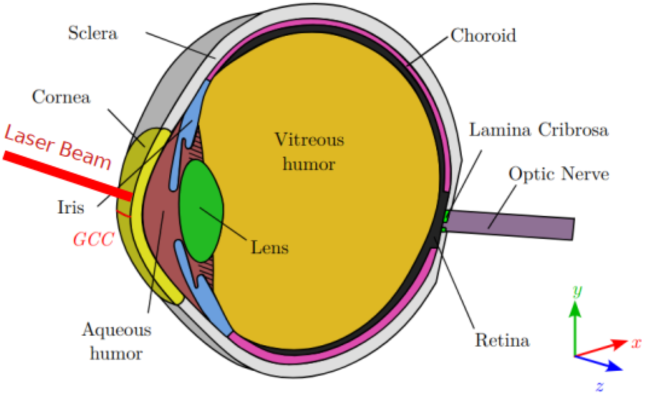


Figure 8: Vertical cut of the geometrical model of the human eye. *["Model order reduction and sensitivity analysis for complex heat transfer simulations inside the human eyeball". Thomas Saigre, Christophe Prud'homme, and Marcela Szopos [SPS24].]*

Eye Geometry: Boundary Conditions

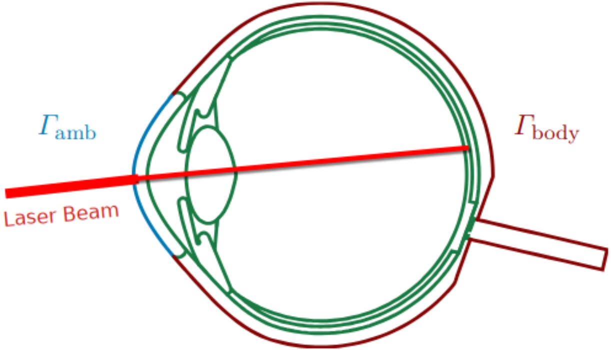


Figure 9: Vertical cut of the geometrical model of the human eye. *["Model order reduction and sensitivity analysis for complex heat transfer simulations inside the human eyeball". Thomas Saigre, Christophe Prud'homme, and Marcela Szopos [SPS24].]*

The Pennes Bioheat Equation

$$
\begin{cases}
\rho_i C_{p_i} \dfrac{\partial T_i}{\partial t} - \nabla \cdot (k_i \nabla T_i) = \boxed{Q_{\mathrm{l},i}} + \boxed{Q_{\mathrm{m},i}} + \boxed{Q_{\mathrm{p},i}} & \text{on } \Omega = \bigcup_{i=1}^{10} \Omega_i, \\
-k_i \nabla T_i \cdot n_i = \underbrace{h_{\mathrm{amb}}(T_i - T_{\mathrm{amb}})}_{(i)} + \underbrace{\sigma\varepsilon(T_i^4 - T_{\mathrm{amb}}^4)}_{(ii)} + \underbrace{E}_{(iii)} & \text{on } \Gamma_{\mathrm{amb}}, \\
-k_i \nabla T_i \cdot n_i = h_{\mathrm{bl}}(T_i - T_{\mathrm{bl}}) & \text{on } \Gamma_{\mathrm{body}}, \\
T_i = T_j & \text{on } \Omega_i \cap \Omega_j, \\
k_i \nabla T_i \cdot n_i = -k_j \nabla T_j \cdot n_j & \text{on } \Omega_i \cap \Omega_j.
\end{cases}
$$

$\boxed{Q_{\mathrm{l},i}} = \mu_{a,i}(\lambda) \times \Phi_\lambda(x, y, z)$: **laser** heat generation,

$\boxed{Q_{\mathrm{m},i}}$: **metabolic** heat generation $\rightarrow$ ignored since negligible compared to the others,

$\boxed{Q_{\mathrm{p},i}} = \rho_{\mathrm{bl}} \omega_{\mathrm{bl}} C_{p_{\mathrm{bl}}} (T_{\mathrm{bl}} - T_i)$: blood **perfusion** heat generation.

The Radiative Transfer Equation (RTE)

Describes the propagation of light in a medium.

$$\left(\frac{1}{c}\frac{\partial}{\partial t} + \boxed{s \cdot \nabla} + \boxed{\mu_a(r) + \mu_s(r)}\right) L(r, s, t) = \boxed{\mu_s(r) \int_{4\pi} L(r, s', t)\beta(s, s')\mathrm{d}s'} + \boxed{Q(r, s, t)},$$

$L(r, s, t)$: **radiance** at position r, in direction s, $(r, s) \in \mathbb{R}^3 \times \mathbb{S}^2$
c: speed of light $[\mathrm{ms}^{-1}]$,
$\beta(s, s')$: phase function [dimensionless].

$\boxed{s \cdot \nabla}$: **advection** — directional transport of light,
$\boxed{\mu_a(r) + \mu_s(r)}$: **attenuation** due to absorption and scattering,
$\boxed{\mu_s(r) \int \dots}$: **in-scattering** contribution,
$\boxed{Q(r, s, t)}$: **source** term.

The Diffusion Approximation

Valid when $\mu_a \ll \mu_s$.

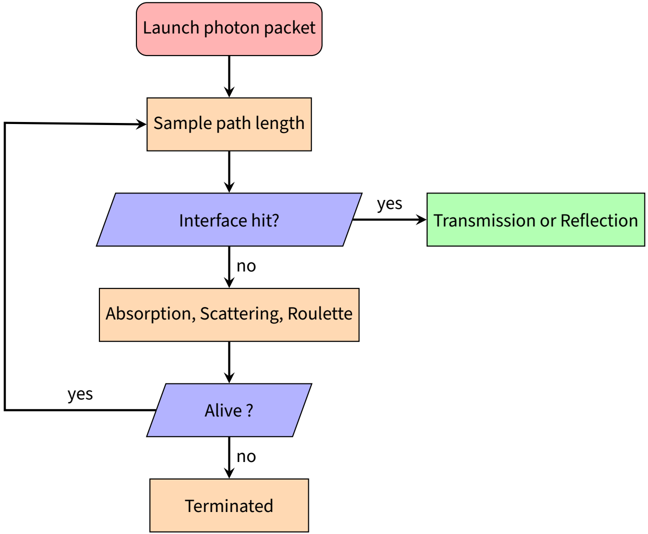
$$\begin{cases} \dfrac{1}{c}\dfrac{\partial \Phi_i}{\partial t} - \nabla \cdot \left(D_i \nabla \Phi_i \right) + \mu_{a,i}(\lambda) \Phi_i = S & \text{on } \Omega = \displaystyle\bigcup_{i=1}^{10} \Omega_i, \\ \Phi_i + 2\, \alpha_i D_i \nabla \Phi_i \cdot n_i = \Phi_{\text{amb}} & \text{on } \Gamma_{\text{amb}}, \\ \Phi_i = \Phi_j & \text{on } \Omega_i \cap \Omega_j. \end{cases}$$

$D_i = \dfrac{1}{3\left(\mu_{a,i}(\lambda) + \mu'_{s,i}(\lambda) \right)}$: diffusion coefficient,

$\alpha_i = \dfrac{1 + R_i}{1 - R_i} \in [0, 1]$: albedo (*i.e.* fraction of light scattered back),

R_i: reflection coefficient.

The MC Approach (RTE-MC)



RTE-MC: Beer-Lambert Law

Using the **Beer-Lambert law**, for a **homogeneous medium** of length x we have:

$$\boxed{N_x = N_0 \, e^{-\mu_t x}}$$

Where:

- $\mathbf{N_0}$: initial number of photons
- $\mathbf{N_x}$: number of photons after traveling distance x
- $\mu_{\mathbf{t}} = \mu_{\mathbf{a}} + \mu_{\mathbf{s}}$: attenuation coefficient, sum of absorption μ_a and scattering μ_s

RTE-MC: Path Length Distribution

The probability density function (PDF) of the photon path length x in a medium is:

$$\boxed{f_x(x) = \underbrace{\mu_t}_{\text{attenuation coeff.}} e^{-\mu_t x}\, \mathbb{1}_{\{x \geq 0\}}}$$

The cumulative distribution function (CDF) is given by:

$$\boxed{F_x(x) = \int_{-\infty}^{x} f_x(t)dt = \int_0^x \mu_t e^{-\mu_t t} dt}$$

Evaluating the integral:

$$F_x(x) = \mu_t \left[-\frac{1}{\mu_t} e^{-\mu_t t} \right]_0^x = 1 - e^{-\mu_t x} \mathbb{1}_{\{x \geq 0\}}$$

Note: $F_x(x)$ is strictly increasing on $\mathbb{R}^+$ and hence invertible, allowing us to sample path lengths from uniform random variables.

RTE-MC: Inverse Transform Sampling

We want to find the **inverse** of the CDF to sample path length x from a uniform random variable $\xi \sim U(0,1)$.

$$\boxed{\begin{aligned} y &= 1 - e^{-\mu_t x}, \quad y \in (0,1) \\ e^{-\mu_t x} &= 1 - y \\ -\mu_t x &= \ln(1-y) \end{aligned}}$$

Since $y \in (0,1)$, define $\xi := 1 - y \in (0,1)$. Then

$$\boxed{s := x = -\frac{\ln(\xi)}{\mu_t}}$$

RTE-MC: Anisotropic Scattering

When the medium is **anisotropic**, the photon travels a distance s in a random direction θ and then scatters.

To sample the **scattering angle** θ, we use the **Henyey-Greenstein** phase function:

$$\cos(\theta) = \begin{cases} 2\xi - 1 & \text{if } g \approx 0 \\ \dfrac{1 + g^2 - \left(\frac{1-g^2}{1-g+2g\xi}\right)^2}{2g} & \text{otherwise} \end{cases}$$

Here, $g \in (-1, 1)$ is the **anisotropy factor**, representing the degree of forward scattering.

RTE-MC: Energy Deposition

At each space step s, the photon (or photon packet) deposits part of its **energy** w (initialized at 1) into the element it lands in.

$$\boxed{w \leftarrow w \cdot \frac{\mu_a}{\mu_t}}$$

Where w is the updated energy of the photon packet after absorption in the current element.

Note: Intermediate elements traversed between steps may not receive any energy deposition unless explicitly handled.

RTE-MC: Energy Map

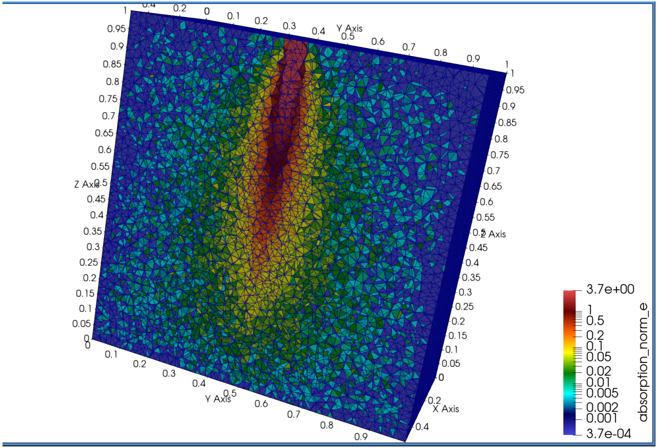


Figure 10: Early results of the map of absorption on a cube.

Bounding Volume Hierarchy (BVH)

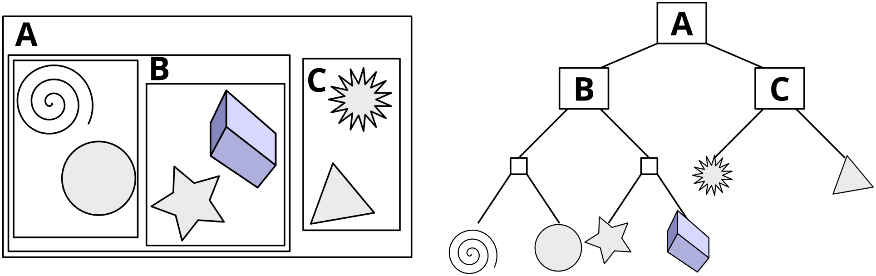


Figure 11: Example of bounding volume hierarchy (BVH) in two dimensions. *[Wikipedia]*

RTE-MC: Fragment-wise Energy Deposition

As the photon travels element-by-element before an interface:

In each segment:

$$\boxed{A_i = w \cdot \frac{\mu_a}{\mu_t} \cdot \left(1 - e^{-\mu_t \cdot \Delta s_i}\right)}$$

- $\mu_t = \mu_a + \mu_s$: total attenuation,
- Δs_i: path length in element i,
- A_i: energy deposited in element i.

Remaining weight:

$$\boxed{w \leftarrow w \cdot e^{-\mu_t \cdot \Delta s_i}}$$

This loop runs until the photon reaches an interface or exhausts its path.

RTE-MC: Tail Absorption After Last Hit

If the ray continues beyond the last known intersection:

Residual segment:

$$\boxed{A_{\text{tail}} = w \cdot \frac{\mu_a}{\mu_t} \cdot \left(1 - e^{-\mu_t \cdot s_{\text{tail}}}\right)}$$

- $s_{\text{tail}} = s - t_{\text{last}}$: remaining distance,
- A_{tail}: energy deposited in tail element,
- $w \leftarrow w \cdot e^{-\mu_t \cdot s_{\text{tail}}}$: update weight.

If the tail lies in a valid mesh element, energy is deposited and the photon advances.

This ensures all traversed regions contribute to absorption, including the “tail” of the step.

RTE-MC: Fresnel Illustration

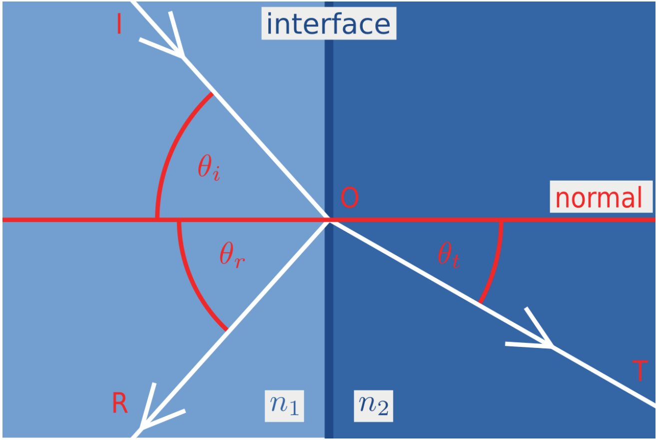


Figure 12: Fresnel equations illustrated ($\theta_i = \theta_r$). *["Fresnel equations" Wikipedia]*

RTE-MC: Interface Interactions

Interface Γ_{ij} hit? reflection ? transmission ?

We rely on:

- **Fresnel equations** to compute the fraction of light that is reflected vs transmitted.
- **Snell's law** to compute the refraction angle:

$$\boxed{n_1 \sin(\theta_i) = n_2 \sin(\theta_t)}$$

- n_1, n_2: refractive indices of the two media,
- θ_i: angle of incidence,
- θ_t: angle of transmission (refraction).

(Snell governs ray direction, Fresnel governs energy splitting.)

RTE-MC: Fresnel Reflection Coefficient

The Fresnel reflection coefficient $\mathbf{R}(\theta_\mathbf{i})$ quantifies the fraction of light reflected at the interface:

$$\boxed{R(\theta_i) = \frac{1}{2}\left[\frac{\sin^2(\theta_i - \theta_t)}{\sin^2(\theta_i + \theta_t)} + \frac{\tan^2(\theta_i - \theta_t)}{\tan^2(\theta_i + \theta_t)}\right]}$$

Note: The reflection depends strongly on the angles θ_i and θ_t, affecting photon path probabilities.

RTE-MC: Schlick's approximation

$$\boxed{R(\theta_i) = R_0 + (1 - R_0)\,(1 - \cos(\theta_i))^5}$$

- $R_0 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$: reflectance at normal incidence,
- θ_i: angle between the incident ray and the surface normal.

We generate a uniform random number $u \sim \mathcal{U}(0, 1)$, and apply:

- If $u < R(\theta_i)$: **reflect** the photon.
- Else: **refract** it using Snell's law (if not internally reflected).

RTE-MC: Russian Roulette

We expect the photon's energy to decrease rapidly. To avoid tracking all low-energy photons while maintaining physical accuracy, we use the **Russian Roulette** technique. That is, if $w < w_{\min}$, then:

Russian Roulette:
Sample a realisation $\xi \sim \mathcal{U}(0,1)$
if $\xi < p$ **then**
  Photon survives: $w \leftarrow \frac{w}{p}$
else
  Photon is terminated
end if

Pennes Bioheat Equation & RTE-MC Coupling

- The RTE-MC provides the **normalized absorption map A**.
- The source term Q_l in the **Pennes equation** is given by:

$$\boxed{Q_l(x, y, z) = P \times A}$$

  where P is the **laser power** and A is the **absorption map** from RTE-MC.
- Solving the **Pennes equation** yields the temperature distribution $T(x, y, z)$ within the eye.

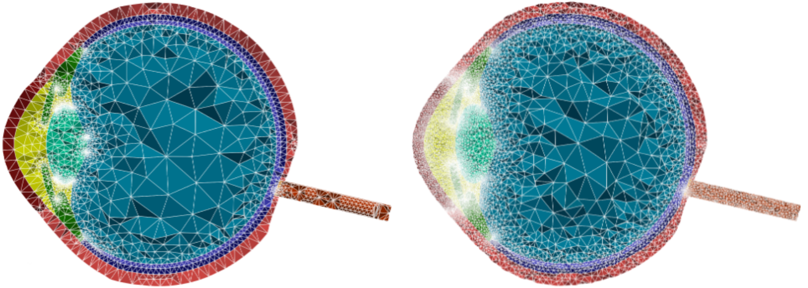
This coupling allows simulation of light-induced heating effects with spatial precision.

Summary: RTE vs Diffusion vs RTE-MC

Aspect	RTE	Diffusion Approx.	RTE-MC
Accuracy	Full angular and spatial resolution	Valid only for highly scattering media	Statistical approximation of full RTE
Complexity	High (5D integro-diff. eq.)	Low (scalar PDE)	Moderate (many photon paths)
Anisotropy	Explicit phase function	Approximate via reduced scattering	Fully supported
Computation	Very costly	Very efficient	Parallelizable
Best use case	Benchmarking, theoretical	Tissue-scale heat estimation	Complex geometry with heterogeneities

Discretization

Space Discretization



(a) Original mesh generated by Salome, with $4.64 \cdot 10^5$ tetrahedrons: $h_{\min} = 5.77 \cdot 10^{-6}, h_{\max} = 5.76 \cdot 10^{-3}$.

(b) Mesh refined around the anterior and posterior chambers, with $9.4 \cdot 10^5$ elements: $h_{\min} = 5.09 \cdot 10^{-5}, h_{\max} = 3.12 \cdot 10^{-3}$.

Figure 13: Meshed geometry of the eye, over a vertical plane. Characteristic sizes are given in meters. *["Model order reduction and sensitivity analysis for complex heat transfer simulations inside the human eyeball". Thomas Saigre, Christophe Prud'homme, and Marcela Szopos [SPS24].]*

Space Discretization: Test Case

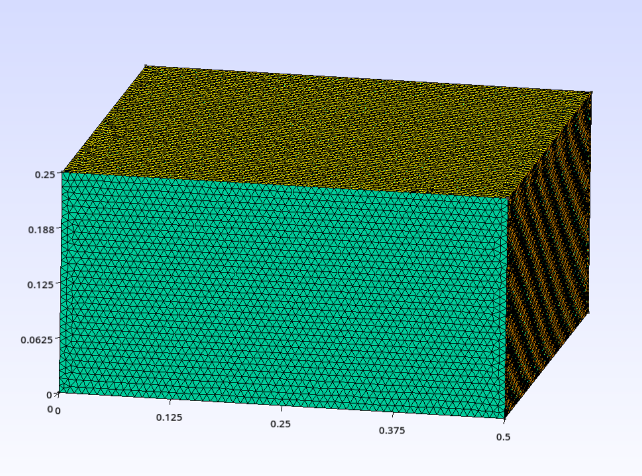


Figure 14: 0.5 x 0.5 x 0.25 cm single layer cube with 81907 nodes and 495013 elements.

Simulations

RTE-MC Simulation Parameters

- **Beam Diameter**: 0.2 cm
- **Absorption Coefficient** μ_a: 5 cm^{-1}
- **Scattering Coefficient** μ_s: 100 cm^{-1}
- **Anisotropy Factor** g: 0.9
- **Number of Photons**: 2,000,000

RTE-MC: Absorption Map

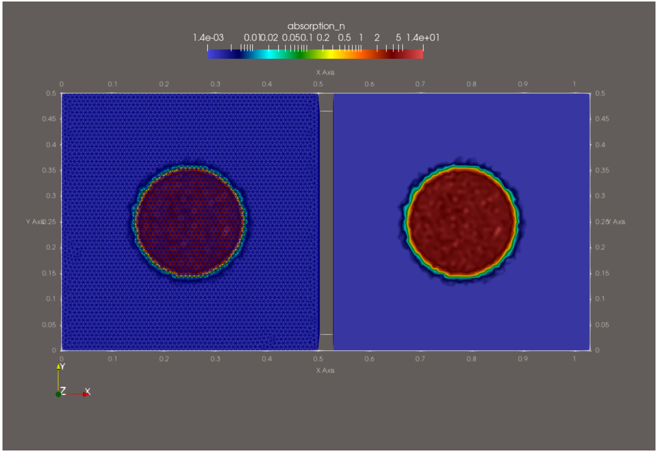


Figure 15: Log scale top view of the absorption map.

RTE-MC: Absorption Map

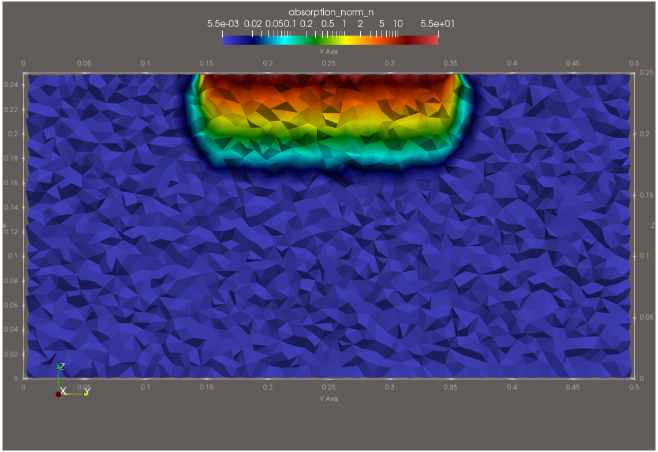


Figure 16: Log scale clipped view of the absorption map for the cube geometry.

Heat Simulation Parameters

- **Laser Power** P: 0.5 W
- **Simulation Time Step** Δt: 0.01 s
- **Total Simulation Time** T: 15 s
- **Time Scheme**: Crank-Nicolson
- **Boundary Conditions**: Homogeneous, Neumann on top, Dirichlet on others
- **Initial Temperature** T_0: 0 °C

Heat Simulation Results

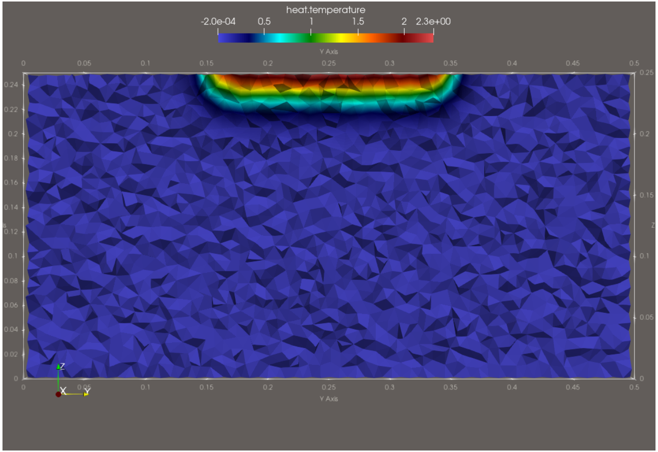


Figure 17: t=0.2 s

Heat Simulation Results

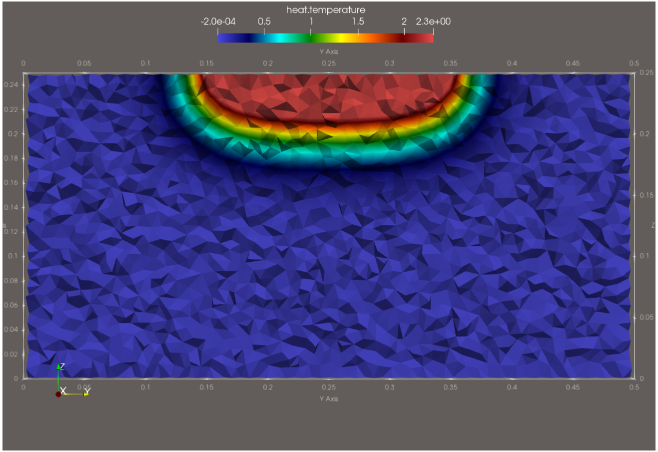


Figure 18: t=1 s

Heat Simulation Results

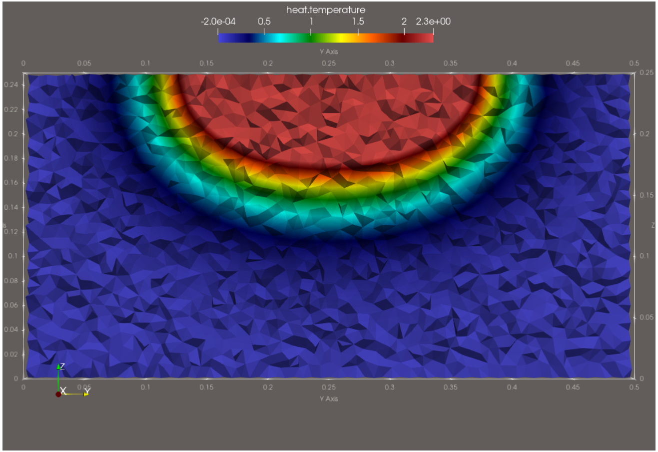


Figure 19: t=0.3 s

Heat Simulation Results

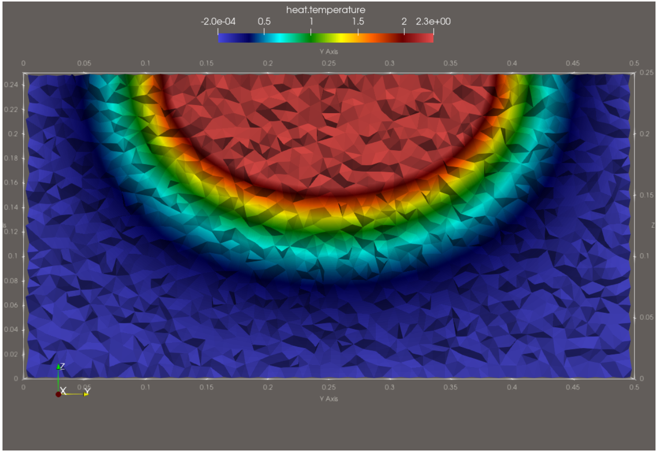


Figure 20: t=0.5 s

Heat Simulation Results

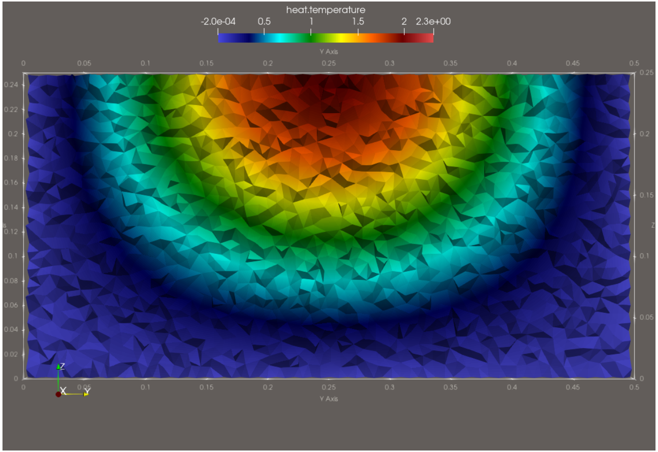


Figure 21: t=1.0 s

Heat Simulation Results

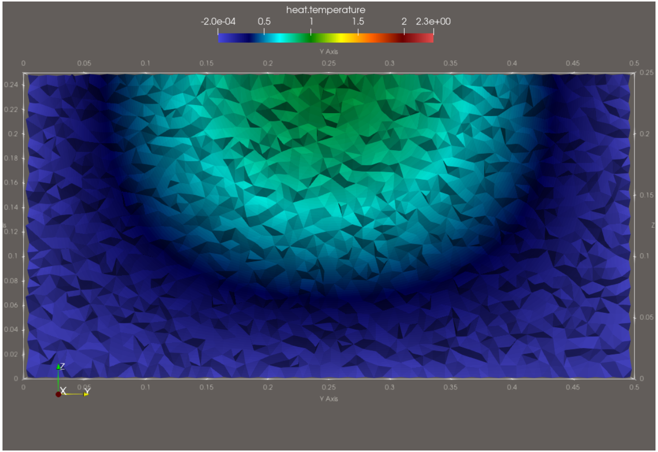


Figure 22: t=15 s

The end

Thank you for your attention!

🎉

Any questions?

Contact Information:

`pierre.antoine.senger@gmail.com`

`github.com/pa-senger`

References

[AF22] Andrea Arnold and Loris Fichera. “Identification of Tissue Optical Properties during Thermal Laser-tissue Interactions: An Ensemble Kalman Filter-Based Approach”. In: *International Journal for Numerical Methods in Biomedical Engineering* 38.4 (Apr. 2022), e3574. ISSN: 2040-7939, 2040-7947. DOI: 10.1002/cnm.3574. URL: https://onlinelibrary.wiley.com/doi/10.1002/cnm.3574.

[Blo19] FindLight Blog. *Laser Tissue Interactions: Biological Factors to Consider for Dermatology*. 2019. URL: https://www.findlight.net/blog/laser-tissue-interactions/.

[SPS24] Thomas Saigre, Christophe Prud’homme, and Marcela Szopos. “Model Order Reduction and Sensitivity Analysis for Complex Heat Transfer Simulations inside the Human Eyeball”. In: *International Journal for Numerical Methods in Biomedical Engineering* 40.11 (Sept. 9, 2024), e3864. ISSN: 2040-7939, 2040-7947.

DOI: 10.1002/cnm.3864. URL:
https://onlinelibrary.wiley.com/doi/10.1002/cnm.3864.

Heat Simulation Validation

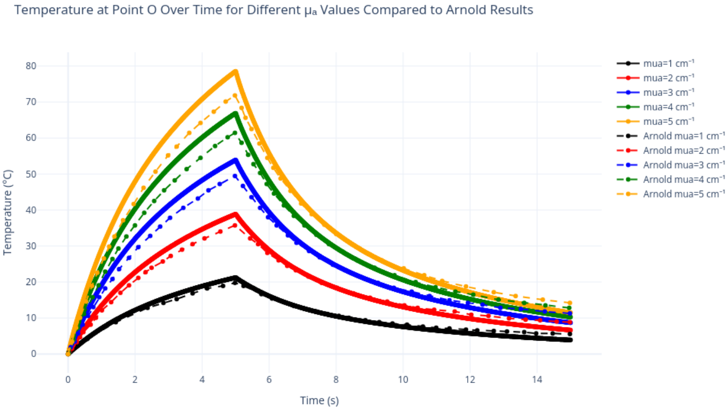


Figure 23: Validation of heat simulation results.

Heat Simulation Validation

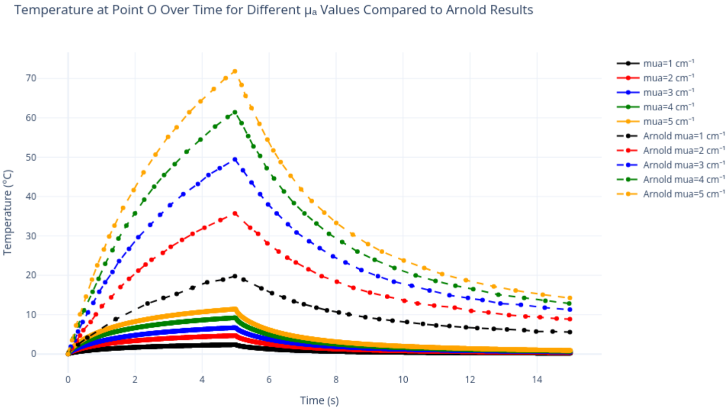


Figure 24: New algo results vs Arnold.